

Analysis of Literature Surrounding Cacao Production for the Purpose of Machine Learning Applications.

Undergraduate Research
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Abstract

Cocoa, a highly valued food source known for its rich flavor and antioxidant properties, faces significant production challenges as global demand rises. This literature review examines interconnected factors affecting Theobroma cacao cultivation, including climate variability, disease prevalence, socioeconomic constraints, and technological barriers. Key insights reveal that optimal growing conditions are increasingly threatened by climate change, while diseases such as frosty pod rot and black pod disease pose serious risks to crop yields. Socioeconomic factors, including an aging farmer population and limited access to education and technology, further exacerbate these challenges. Despite the potential of machine learning and deep learning technologies to enhance disease identification and improve agricultural practices, their implementation in rural farming contexts remains limited. This paper advocates for a shift towards more accessible, traditional machine learning methods that could empower farmers and improve cocoa production. The findings highlight the need for strategic investment in education, technology, and management practices to ensure the sustainability of the cocoa industry. Future research should focus on developing practical, low-cost technological solutions tailored to the unique conditions of cocoa farming, emphasizing decentralized implementations that can deliver significant benefits to rural communities.

Keywords: Cocoa, Cacao, Machine Learning, Climate.

I. Introduction

Humanity has long valued Cocoa as a delicious and antioxidant-rich food source. Since its spread to Europe during the Columbian Exchange, year-over-year demand for this sweet treat has steadily grown; in an effort to keep pace with demand, *Theobroma Cacao* is presently grown in virtually every tropical climate, but the majority of output is outside its native South America.

Relevant literature sources reveal several interconnected factors, including the question of climate variability, disease, socioeconomic, and technological challenges, which are stressing current production efforts. In light of these challenges, this paper seeks to highlight key insights from these sources in hope of inspiring current and future Machine Learning research. I believe this topic in particular needs the exact opposite of big data and massive compute models; both rural farmers and the complex cocoa industry may find unexpected benefit from traditional Machine Learning methods.

II. Literature

a. Climate Factors

Cacao favors a relatively narrow range of tropical conditions. These conditions, as we'll see, lead to inherent issues. Optimal conditions vary throughout the growth cycle of the plant; pollen germination is optimal within 25 – 30 C; flower production is optimal at 26.7 – 30 C; optimal growth prefers 18.6 – 20.8 C. This metric for growth has a high degree of confidence, given similar research has been conducted in greenhouses, where the plant reacted favorably to very similar conditions [Climate Change and Cacao Reproduction]. Similarly, rainfall and humidity are important factors; the tropics are subject to wet and dry seasons, which at their extremes can massively affect leaf production, blooming, and seed pod production. [Cacao Ecophysiology].

Given that the plant takes several years to mature, and the pods are slow to reach maturity, non-ideal conditions such as these can have a severe and long-lasting impact, compared to less complex flora. As this author notes, the warming trend in Africa and extreme weather are expected to continue for the foreseeable future [Cacao Agroforestry]. Given the Ivory Coast is currently the primary producer of Cacao [V.C. Cacao Infographic], climate is a key point of interest for research.

b. Disease

Several diseases affect Cacao in various ways. *Moniliophthora roreri* causes “frosty pod rot,” which is particularly problematic for rural farmers, as

it can't be easily identified visually, and hence is not readily treatable [Socioeconomic and Technical Factors of Cacao Farming]. In their research, Barrera also notes a similar disease, "black pod disease," which *is* easily spotted with the naked eye, but equally as detrimental to crop production. Deep learning has shown promise in identifying both of these afflictions; interestingly, the research seems to suggest that even monolilia can be spotted with some degree of accuracy, taking into account visual data from the leaves, stems, and slight variations in the pods which might go unnoticed by the human eye [Deep Learning Model for Cacao Disease Identification].

In Barrera's research, we see that mitigation, or "cultural methods," are quite primitive. Given the disease is primarily fungal, clean tools, removing rotten pods or leaves, and looking for visible signs of fungus growth on the trees are the main methods used by rural farmers [Socioeconomic and Technical Factors of Cacao Farming]. In support of the case for incorporating Machine Learning or deep learning, there is an obvious connection: two major sources of disease are able to spread unseen to the naked eye. Perhaps other less obvious changes to the plant can indicate the presence of fungus before the visible colony is detected by humans. Anyone familiar with mold and fungus understands that visible fungal growth comes long after it's already well-established. These primitive mitigation methods against fungus are clearly insufficient.

This conclusion seems to be long-standing, as work to develop such a computer-driven aid began as early as 1996, with research into "expert systems" consisting of a Graphical User Interface, Inference Engine, and a Knowledge Base (driven by experts behind the scene). This system would essentially take in various parameters about the plants (via the GUI) and use if-then rules (a primitive decision tree model), and return recommendations and corresponding uncertainty values [Expert System for Cacao Disease Diagnosis]. Hence, we see that this research is not new, but obviously attempts to deploy such technology to rural areas have been unsuccessful for obvious economic and practical reasons.

c. Socioeconomic Factors

Age, education, and labor issues are at the forefront of the many socioeconomic factors impacting Cacao farmers. Most farmers of this crop are over 50, uneducated, and are essentially subsistence farmers. The work is hard and the return for the effort is marginal. Typically, such farming efforts are non-commercial and typically are a family affair. The sources note that there's an ongoing decrease in the number of individuals willing to grow these crops [Cacao Agroforestry].

Another challenge is the lack of technical assistance, which was previously explored as a major aid in disease identification. A combination of a gen-

eral lack of technology in underdeveloped regions, a lack of skilled computer technicians and programmers, the difficulty of supplying power and transporting bulky computer equipment in rainy growing seasons, and a general adversity to technology [Socioeconomic and Technical Factors of Cacao Farming], are all factors which need to be considered carefully when considering applying technology-intensive methods like Machine Learning to this problem.

Perhaps of larger consequence is a general lack of organization throughout these underdeveloped regions. Land ownership is often informal and poorly managed, and violent disputes over croplands are common [Socioeconomic and Technical Factors of Cacao Farming]. Barrera also aptly points out that loans are generally not an option, which implies if a technical option were to be viable, it would need to be of marginal cost with minimal expense in upkeep.

d. Technological Challenges

Genetics are of particular interest to the future viability of successful Cacao operations. This is one notable area where technological innovation is already in place and has been relatively successful in mitigating losses. With knowledge of plant strains and breed characteristics, improved pollination, disease tolerance, productivity, and compatibility with existing agroforestry setups can be taken into consideration [Socioeconomic and Technical Factors of Cacao Farming]. Likewise, Barrera notes that soil testing, proper timing (factoring in weather predictions), well-managed irrigation, and access to general technical assistance shows a lot of promise in abating the industry’s current struggles where growing is concerned.

However, the struggles persist beyond the growth stage. Cacao pods need to ferment properly, wherein the color changes and the flavor develops prior to being dried. Surveys and inspections of farms indicate that often this critical step is poorly managed, which indicates that the issues run beyond what we’ve explored this far. Poorly or outright improper fermentation beds are common, which inhibit the pods from drying properly, resulting in huge amounts of waste [Socioeconomic and Technical Factors of Cacao Farming]. Barrera also observes that harvesting is often done with improper tools, which damage the trees unnecessarily, slowing future growth. In line with aforementioned education and economic deficiencies, these practices are somewhat understandable.

III. Conclusion

a. Summary of Findings

In summary, it has been seen that the issues facing Cacao production are varied, though essentially interconnected. Farmers are facing economic issues, climate issues, labor issues, and are often not educated in growing these complicated plants beyond their own first-hand knowledge. Their prospects and outlook are frankly rather bleak. Given that production is limited to these climate regions, these issues are likely not going to change overnight, which is unfortunately what the industry seems to need if production is going to continue to meet long-term demand.

b. Implications

As we've seen, technology does show promise when it is implemented in a reasonable way. When deployed broadly, technology is useful for selecting appropriate genetics for the farmers to select for growing new crops, or grafting to existing trees. This is promising, as it shows the industry is already attempting to solve its problems, albeit in a somewhat detached manner.

It's encouraging that there are past and current efforts to improve crop yields through computer-controlled irrigation, and early disease identification via deep learning. This follows the pattern that many agricultural areas worldwide are currently benefiting from – wherein computer technology is an aid to farmers, which addresses the labor shortages and economic shifts that the rest of the developed world has seemingly overcome. The implication is that the Cacao industry presently requires strategic investment, education, and proper management.

c. Research Gaps

It seems very little has been done to implement these theoretical solutions (ML, Deep Learning) into actual operations. It seems obvious that until such efforts are made in earnest, the industry will continue to lose out on any benefits therein. Theory and any advancement will likely be of little use when the technology isn't being developed according to actual conditions, nor implemented in the crops.

With this in mind, particular considerations should be made, such as total cost, power constraints, internet connectivity requirements, maintenance, knowledge required to implement and operate, and so on. In short, simple and effective solutions seem likely to have the biggest impact. AI-centric, power-intensive, big data solutions seem very unlikely to be practical, which goes directly against the current trend in 2025.

d. Future Directions:

All things considered, it seems that the Machine Learning has potential to be impactful in this industry. A well-trained model could predict fungal infections before they're visible on the trees; perhaps drones with cameras could achieve this end. Smarter irrigation which can take into account localized weather predictions could conserve water and lend to better yields. Likewise, smart drying beds could result in less losses on otherwise viable yields.

Ultimately, I believe there are many unexplored cases where simple technology powered by effective machine learning algorithms could be of great benefit. In particular, I propose that self-contained, robust, decentralized implementations would seem to logically be most promising for having the greatest impact. Presently, further research and surveys seems necessary in order to narrow down on an ideal pilot project.

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